

National Taipei University of Technology

Object-Oriented Analysis and Design

Homework #7

This is the last homework of this semester. You are required to implement the use cases chosen in homework 6. The implementation activities include domain model programming (writing production code) and unit testing (using xUnit framework). That is, in this homework you should provide source code listing and testing code listing. Snapshots, such as the execution of your system and the result of unit testing, may be provided as well. You also have to construct the implementation class model, and show the difference between the new implementation class model and the design class model you made. The structure of this homework is basically the same as that of homework 5. In order to make grading easier, **please mark the new material written for this homework as “new” for each part of the homework.** For example, if you have a new class in the design class diagram or a new sequence diagram, mark them new; mark any new code, new classes, new methods, and new subsections clearly so that they can be easily identified.

1 Cover Page

The cover page includes the *project title*, *homework number*, and *team members*.

2 [5%] Requirement Document

2.1 Change History

2.2 Problem Statement

2.3 Summary of System Features [Larman04, Ch7.7]

2.4 Use Case Diagram [Larman04, Ch6.17]

2.5 Use Cases [Larman04, Ch6]

2.6 Non-functional Requirements and Constraints [Larman04, Ch7.4]

2.7 Glossary [Larman04, Ch7.9]

2.8 Software Environments (The original item of development language in homework 1)

3 [5%] Domain model

3.1 Domain class diagram showing only concepts [Larman04, Ch 9.4 ~ 9.6]

3.2 Add associations [Larman04, Ch 9.14]

3.3 Add attributes [Larman04, Ch 9.16]

4 [15%] Design

4.1 Draw a Logical Architecture with UML package diagram [Larman04, Ch 13]

4.2 Use-Case Realizations with GRASP Patterns [Larman04, Ch 17]

4.2.1 Choose the most significant use-case, identify system events, and draw a system sequence diagram for that use case.

4.2.2 Draw a sequence diagram for each system event you identified, and label with the GRASP patterns to describe how you assign responsibilities.

4.3 Design Class Model (domain class model after class design) [Larman04, Ch 16]

5 [15%] Implementation Class Model

5.1 Draw an implementation class diagram for your system (including associations, attributes and methods)

5.2 Show the difference between implementation class model and design class model

Construct the following two tables:

Table 5.2.1: Comparison with design and implementation class

Class	Method	Design	Imp.
C1	m1	Yes	Yes
	m2	No	Yes
C2 (C2 is added in design)	m1	No	Yes
	m2	No	Yes
C3 (C3 is removed in design)	m1	Yes	No
	m2	Yes	No

Table 5.2.2: Summary of implementation class/method changed

	Number of added	Number of removed	Number of modified
Class			
Method			

Note: Number of modified means the number of modification of signature.

5.3 Calculate Line of Code

Provide the following table for your implementation classes:

Table 5.3.1: Line of Code of classes

No	Class Name	Number of methods	Line of Code in Class (without comment)
1			
2			
...			
Sum			

6 [15%] Programming

6.1 Snapshots of system execution

- 6.2 Source Code Listing (note: select and list the most important source code of both iterations; no more than 10 pages)
- 7 [15%] Unit Testing
 - 7.1 Snapshots of testing result
 - 7.2 Unit Testing Code Listing (note: select and list the most important test code of both iterations; no more than 10 pages)
- 8 [30%] Revise your requirement document, domain model, design, implementation, programming, and unit testing based on TA's comments or your own findings, if any. In case that the grading of HW#6 is not completed yet, you should contact TA and obtain comments before submitting this assignment.

9 Measurement

How many hours did you spend in this homework? Please record the time precisely in “YY/MM/DD HH:mm ~ HH:mm” format. **Keep logging precisely for each team member.**